
PART B

Human-esque

Implications of Robot Anthropomorphism

There is an universal tendency among mankind to conceive all beings like themselves, and to transfer to every object, those qualities, with which they are familiarly acquainted, and of which they are intimately conscious. We find human faces in the moon, armies in the clouds; and by a natural propensity, if not corrected by experience and reflection, ascribe malice or good-will to everything, that hurts or pleases us.

—David Hume, *The Natural History of Religion**

* David Hume, *The Natural History of Religion* (J J Tournaisien 1793) 12.

Chapter IV: Challenges of Robot Anthropomorphism

Actions such as his could come only from a robot, or from a very honourable and decent human being. But you see, you just can't differentiate between a robot and the very best of humans.

—Isaac Asimov, *I, Robot*¹

Introduction

During the discussion of the Genesis creation account in the last chapter, it was concluded that Judeo-Christian theology considers self-management capacity as one of the core characteristics that distinguish human beings from other creatures and render human beings somehow similar to God. It is thought that the Genesis creation account reveals that self-management capacity is one of the more exclusively 'human' attributes. Naturally, the mere fact that robots -which are ultimately functional artefacts – are relatively autonomous or can display the capacity for self-management encourages human beings to attribute other human characteristics to them.² In short, the capabilities associated with the characteristic of relative autonomy enable robots to prompt significant levels of anthropomorphic responses from human beings.

Anthropomorphism is not some new phenomenon that is just emerging with the proliferation of robots. It is thought to be one of the oldest survival mechanisms of human beings, developed at the earlier stages of human evolution, helping the first human beings to stay alert in the wild.³ If human beings suppose that movements they feel or sounds they hear in nature belong to possibly hostile human beings or wild animals, they are bound to be more reluctant to let their guards down.⁴ Then, it can be suggested that human beings' anthropomorphic responses toward non-human entities are not exclusively prompted because of the display of self-management capacity by these entities. In that context, regardless of whether the given non-human entity can self-manage, that entity can elicit anthropomorphic responses from human beings so long as there is some display of human-like attributes.⁵

¹ Isaac Asimov, *I, Robot* (first published 1967, HarperVoyager 2018) 204.

² Marco Aurélio Castro, *Robotic Law: The Legal Personality of the Robot* (Kindle edn, 2019) ch 3, s 3.5 sub-s 5.2.4.

³ Marco A C Varela, 'The Biology and Evolution of the Three Psychological Tendencies to Anthropomorphize Biology and Evolution' (2018) 9 *Frontiers in Psychology*.

⁴ Mike Dacey, 'Anthropomorphism as Cognitive Bias' (2017) 84 *Philosophy of Science* 1152, 1152-1154.

⁵ Cédric Sueur, Marie-Amélie Forin-Wiart, and Marie Pelé, 'Are They Really Trying To Save Their Buddy? The Anthropomorphism of Animal Epimeletic Behaviours' (2020) 10 *Animals* 2323, 2-3.

Ordinarily, human beings do not wilfully decide to anthropomorphise non-human entities following mindful thinking processes. Instead, anthropomorphic responses are often reflexive and bypass the conscious evaluation procedures that perceptions would typically be subjected to before any voluntary reaction. The extent and strength of anthropomorphic responses by a non-human entity are proportional to the degree to which that non-human entity displays attributes and traits considered human.⁶ In other words, the intensity of anthropomorphism is determined by how 'human' is the entity in question.

Although anthropomorphism is not restricted to human-robot interactions, anthropomorphic responses invoked in such interactions can be substantial even concerning non-social robots since they display the capacity to self-manage, regarded as one of the most exclusively human attributes. Anthropomorphic responses elicited by social robots can be expected to be even more robust and last longer since social robots are also equipped with the additional capabilities associated with the characteristic of social agency. Since anthropomorphism involves the intuitive perception of non-human entities as similar to human beings, the phenomenon inevitably influences how human beings behave in social contexts.⁷ Social robots are readily perceived as social actors because of the significant levels anthropomorphic responses they invoke.

The law, in essence, regulates the social behaviours of human beings to fulfil its functions. Consequently, the present study admits that anthropomorphic responses that robots -especially social robots- invoke, and the impacts of such anthropomorphic responses on human beings' social behaviours can pose unique challenges to the law's function. In that respect, the following chapter explores the legal challenges arising from the phenomenon of robot anthropomorphism.

The chapter is organised into three sections. In the first section, the phenomenon of robot anthropomorphism is introduced, and the variables that affect the phenomenon's intensity are described by utilising various psychological experiments. The second section focuses on undesirable outcomes of the phenomenon and discusses whether any of these outcomes relate to the legal values that underlie existing legal instruments. Finally, the third section examines whether the extent and strength of robot anthropomorphism justify any legal protection being extended to at least some robots.

⁶ Gabriella Airenti, 'The Development of Anthropomorphism in Interaction: Intersubjectivity, Imagination, and Theory Of Mind' (2018) 9 *Frontiers in Psychology*.

⁷ Lingyao Yuan and Alan R Dennis, 'Acting Like Humans? Anthropomorphism and Consumer's Willingness to Pay in Electronic Commerce' (2019) 36 *Journal of Management Information Systems* 450, 454.

IV.1 Robot Anthropomorphism

The following section describes the phenomenon of anthropomorphism and explores the factors that affect the phenomenon's occurrence concerning robots. The section discusses the phenomenon's relationship with the distinctive characteristics of social robots and provides the primary concepts necessary for developing an in-depth understanding of the legal significance of the phenomenon. To that end, the section is organised into two sub-sections. The first sub-section provides general explanations using the available anecdotal and theoretical evidence to establish basic rational connections between the phenomenon and the distinctive characteristics of social robots. The second sub-section supports these general explanations through the findings of various psychological experiments on the extent and strength of anthropomorphic responses invoked by social robots.

IV.1.1 Recorded Experiences

The phenomenon of anthropomorphism refers to the attribution of human characteristics, such as emotions, intentions, sentiments, and even intelligence, to non-human entities. Non-human entities that can invoke anthropomorphic responses include living entities but are in no way limited to them: natural objects or human-made artefacts can also prompt the occurrence of the phenomenon.⁸ Human beings tend to anthropomorphise specific non-human entities intuitively and reflexively, without conscious thought. In other words, anthropomorphic responses are automatically prompted when human beings encounter human-like attributes or behaviours in non-human entities that they cannot immediately and intuitively understand using the knowledge they have at hand.⁹ Non-human entities need to have neither any humanoid physical attributes nor even any form of embodiment to prompt anthropomorphic responses, even non-physical software entities can be anthropomorphised so long as they display some human-like attributes, such as some social interaction skills.¹⁰

⁸ Esmeralda G Urquiza-Haas and Kurt Kotrschal, 'The Mind Behind Anthropomorphic Thinking: Attribution of Mental States to Other Species' (2015) 109 *Animal Behaviour* 167, 168.

⁹ *ibid* 172.

¹⁰ Human beings also anthropomorphise virtual objects: 'In the video game Portal, for example, when players are required to incinerate the companion cube that has accompanied them throughout the game, some will opt to sacrifice themselves rather than the cube, forfeiting their victory.' Kate Darling, 'Extending Legal Protection to Social Robots: The Effects of Anthropomorphism, Empathy, and Violent Behavior Towards Robotic Objects' in Ryan Calo, A Michael Fromkin and Ian Kerr (eds), *Robot Law* (Edward Elgar 2016) 216.

The phenomenon's occurrence regarding non-physical entities is not new, and its intuitive occurrence toward non-physical software entities dates back to the 1960s when human beings who interacted with it were recorded to anthropomorphise ELIZA, the psychotherapist software. That software's only human-like attributes were the natural language texts that it was able to display, and it did not even have the capabilities of relative autonomy.¹¹

Anthropomorphism is not binary but instead incremental. The extent and strength of anthropomorphic responses prompted by non-human entities are proportional to the degree that these entities emulate human beings' behaviours or have human-like attributes. For example, non-human entities with human-like physical features (such as tree barks that are shaped like human faces) invoke weaker anthropomorphic responses than those that can emulate human beings' communication and social interaction skills. The capabilities that are shared by all robots, those associated with the characteristic of relative autonomy, already invoke significant levels of anthropomorphic responses from human beings on their own, as the recorded experiences and various psychological experiments explored below to confirm. However, the anthropomorphic responses prompted in that manner are often incidental and ephemeral. Since non-social robots are not intended to actively and continuously prompt anthropomorphism, anthropomorphism invoked by these robots often discontinues or weakens once human beings observe them or interact with them closer. Unlike non-social robots and because of the capabilities associated with relative autonomy and social agency in social robots, they are expected to prompt longer-lasting and more robust anthropomorphic responses. That is intentional, and by design, the additional capabilities associated with the characteristic of social agency are devised to prompt the sustained occurrence of anthropomorphic reactions.

Moreover, eliciting anthropomorphic responses appears crucial for social robots to fulfil most of their specified functions, especially regarding their roles in the education and health sectors. In these sectors, social robots are currently utilised to facilitate communication between patients and their doctors, engage children with special needs in learning activities, and help vulnerable adults by interacting with them or by motivating, monitoring, and coaching them.¹²

¹¹ Gabriella Airenti, 'The Cognitive Bases of Anthropomorphism: From Relatedness to Empathy' (2015) 7 *International Journal of Social Robotics* 117, 123.

¹² Cory D Kidd, Will Taggart, and Sherry Turkle, 'A Sociable Robot to Encourage Social Interaction Among the Elderly' (IEEE International Conference on Robotics and Automation, Orlando, May 2006); Evan Ackerman, 'MIT's DragonBot Evolving to Better Teach Kids' (*IEEE Spectrum*, 16 March 2015)

Though continuous anthropomorphism is the intentional result of social robots' design features, there might be some undesirable outcomes associated with that degree of anthropomorphism, including the loss of authenticity in human relationships, new types of threats to information security and privacy, potential normalisation of violent behaviours against human beings and animals because of the impunity of such behaviours toward social robots, and the potential exploitation of human beings' emotional attachments to social robots.¹³

The present study first examines the association between the phenomenon of anthropomorphism and the capabilities associated with the characteristic of relative autonomy. The recorded experiences documenting human beings' anthropomorphic responses towards various non-social robots illustrate the extent and strength of anthropomorphic responses invoked just by the capabilities of relative autonomy. For instance, the Roomba, the world-famous brand of robotic vacuum cleaners, are flat, round robots with simple algorithms allowing them to clean floors independently and have none of the capabilities associated with the characteristic of social agency whatsoever. The experiences concerning human beings' approaches to the Roomba point out that the mere fact that these robotic vacuum cleaners can move around on their own prompted the generation of substantial anthropomorphic responses toward them:

Some will clean for the Roomba, so that it can get a rest, while others will introduce their Roomba to their parents, or bring it along when they travel because they managed to develop a (...) relationship: 'I can't imagine not having him any longer. He's my BABY.'¹⁴

As the Roomba are different from other non-robotic vacuum cleaners in the market just because they can self-manage, these recorded experiences establish that the capabilities of relative autonomy are sufficient for human beings to intuitively attribute human sentiments to the Roomba at such substantial levels that they even form emotional attachments toward the Roomba, even though manufacturers did not intend these robotic vacuum cleaners to invoke anthropomorphic responses.

Furthermore, considering that the specified functions of the Roomba are related to cleaning floors, anthropomorphic responses developed toward them to the degree that human beings are willing to clean for the Roomba even appear to impede the utility of acquiring these robotic vacuum cleaners.

¹³ Kate Darling, 'Who's Johnny? Anthropomorphic Framing in Human–Robot Interaction, Integration, and Policy' in Patrick Lin, Ryan Jenkins and Keith Abney (eds), *Robot Ethics 2.0: From Autonomous Cars to Artificial Intelligence* (OUP 2017) 175.

¹⁴ *ibid.*

Apart from the loss of utility, the evocation of anthropomorphic responses by non-social robots does not appear to have any significant adverse outcomes on human beings' legally meaningful personal interests in the case of the Roomba. Still, it is thought that life-threatening situations may arise when non-social robots are deployed to help or replace human beings in the performance of dangerous tasks. In that respect, in the literature, several authors have documented the recorded experiences of soldiers who utilise bomb-defusing and mine-cleaning robots on battlefields. Again because of the capabilities of relative autonomy, soldiers who utilised these robots first tended to project human emotions and sentiments to them and then develop emotional attachments toward them. Anthropomorphic responses invoked in these contexts were so strong that some soldiers were willing to risk injury to rescue these non-social robots they saw as comrades-in-arms.¹⁵

Social robots, in addition to the relative autonomy shared by all robots, are also equipped with specific capabilities associated with the characteristic of social agency. These additional capabilities enable social robots to display socially indicative signals that human beings automatically and subconsciously associate with human emotional and mental states. In short, thanks to the characteristic of social agency, social robots can actively engage with the ingrained anthropomorphic tendencies of human beings with whom they enter into interactions instead of incidentally and ephemerally eliciting anthropomorphic responses like other non-human entities. Some algorithms in the market today have natural language processing capabilities that are so sophisticated that these algorithms cannot be distinguished from human beings in everyday conversations. Moreover, most social robots can recognise human emotions, display meaningful emotional expressions, and perform behaviours that human beings consider to be coherent, intentional, responsive, and emotionally appropriate.¹⁶ Sherry Turkle describes the social interactivity facilitated by the additional capabilities of social robots as follows:

Computers no longer wait for human beings to project meaning onto them. Now, sociable robots meet our gaze, speak to us, and learn to recognise us. They ask us to take care of them; in response, we imagine that they might care for us in return.¹⁷

¹⁵ Doree Armstrong, 'Emotional Attachment To Robots Could Affect Outcome On Battlefield' (*UW News*, 17 September 2013)

¹⁶ Darling, 'Extending Legal Protection to Social Robots' (n 10) 218; Kerstin Dautenhahn and others, 'Socially Intelligent Agents: Creating Relationships with Computers and Robots' in Kerstin Dautenhahn and others (eds), *Socially Intelligent Agents* (Kluwer 2002) 6; Olivia Solon, 'Google's Robot Assistant Now Makes Eerily Lifelike Phone calls for you' (*The Guardian*, 8 May 2018)

¹⁷ Sherry Turkle, *Alone Together: Why We Expect More from Technology and Less from Each Other* (Basic 2012) 2.

Turkle's description of social agency indicates that through the additional capabilities associated with that characteristic, social robots are intended to reach out to human beings at personal, social, and eventually at emotional levels. To put it differently, the characteristic of social agency enables social robots to actively seek anthropomorphic responses, instead of being passive subjects of human beings' projection of human characteristics. The success of these additional capabilities in invoking anthropomorphic responses appears to have been borne out in the recorded experiences related to social robots and is further confirmed by various psychological experiments.

One example of such recorded experiences is the public outrage against one of the videos released by Boston Dynamics. The video showed the company's doglike robot 'Spot' being kicked by human beings to demonstrate the robot's stability.¹⁸ At this point, it must be noted that 'Spot' was not designed to fulfil any social functions. However, because of its zoomorphic physical attributes, as well as its ability to navigate in its environment independently and perform some tasks autonomously, along with some basic communication skills, it is evaluated that it can be regarded as one of the social robots. In any case, Spot's features prompted strong anthropomorphic responses from human beings who viewed the video online.¹⁹ Soon after the video was released, many on the Internet expressed their discomfort over what they viewed as the maltreatment of some form of new sentient entity. The online reaction was so strong that even the global animal rights organisation PETA was forced to acknowledge the incident.²⁰

Some other recorded experiences related to anthropomorphic responses invoked by social robots, compiled by Darling, support the conclusions regarding responses to the supposed maltreatment of 'Spot'. These experiences are related to robots that actually have specified functions of social nature.

Owners of Sony AIBO dogs in the 1990s, while fully aware that it was a robot, would regularly ascribe lifelike essences and mental states to their artificial companion. (...) They would remove their AIBO from the room while changing clothes, so that they would not be "watched," or that they experienced feelings of guilt when putting the device back in its box.

Students in MIT's Media Lab would often put up a curtain between themselves and Kismet, a social robot that simulates emotion through facial expressions, because the lifelike behavior of the face distracted them. And Cynthia Breazeal, Kismet's developer, reports experiencing "a sharp sense of loss" when she parted ways with her own creation at the end of her dissertation.²¹

¹⁸ Christoph Bartneck and Merel Keijsers, 'The Morality of Abusing A Robot' (2020) 11 Paladyn, Journal of Behavioral Robotics 271, 273.

¹⁹ *ibid.*

²⁰ Phoebe Parke, 'Is It Cruel to Kick a Robot?' (CNN, 13 February 2015),

²¹ Darling, 'Extending Legal Protection to Social Robots' (n 10) 217-220.

According to these recorded experiences, human beings' anthropomorphic responses are stronger and more continuous when it comes to robots with specific capabilities associated with the characteristic social agency. While human beings formed significant emotional attachments to non-social robots that they interacted with, these attachments and their anthropomorphic responses did not go as far as feeling like 'they are being watched' by non-social robots or experiencing 'guilt' when they switched them off.

Before the experiments on robot anthropomorphism are elaborated, it is worth mentioning that the phenomenon of anthropomorphism is often explored in popular culture as well, specifically in the context of human beings' reluctance to switch off the non-human entities that they relate to as fellow members of society. In her book *Personhood in Science Fiction*, Juli Gittinger cites the TV series *The Good Place* as one example of exploring anthropomorphism through the medium of fiction.²² The non-human entity that invokes anthropomorphic responses in the series is named Janet, and though she looks indistinguishable from human beings, she is produced to serve as the operational mainframe for one of the afterlife neighbourhoods.

In one episode of the show, during their effort to prevent Janet from taking one of their friends to the Bad Place (the Inferno), Eleanor and Chidi, two of the show's main characters, decide to terminate Janet's program.²³ Janet, programmed to accommodate every wish of the neighbourhood residents, takes them happily to the beach, where her kill switch sticks out of the sand:

JANET: Here we are (*gesturing to a large stand with a red button on it*). Just press this and it's goodbye Janet. (*Chidi groans and frowns*) Chidi, I can see that you're worried. And I just want to assure you—I am not human, and I cannot feel pain.²⁴

However, it is revealed that Janet is also programmed with a failsafe to plead for her life. As Chidi and Eleanor approach the stand with the red button on it, Janet starts screaming, and when they back out, she returns to her original cheerful demeanour, reminding the main characters that she is 'merely an anthropomorphised vessel of knowledge, designed to make their after-lives easier. As the characters try, again and again, to press the kill switch, Janet's pleading goes on with increasing authenticity, and not even Eleanor—the least morally concerned character on the show —can switch Janet off.

²² Juli L Gittinger, *Personhood In Science Fiction* (Palgrave Macmillan 2019) 116.

²³ *ibid.*

²⁴ Michael Schur, *The Good Place - Chapter 7: The Eternal Shriek* (NBC 2016).

Though *The Good Place* explores the extent of anthropomorphism concerning human beings' inability to terminate robots, there are more profound explorations of the phenomenon of anthropomorphism in the cult TV series *Star Trek*. In one of the iconic episodes of *Star Trek: The Next Generation*, *The Measure of a Man*, the only android (humanoid robot) officer of the USS Enterprise, Lieutenant Commander Data, is requested to report to be deactivated and disassembled by an ambitious Federation scientist who hopes to gain enough technical knowledge to construct more androids.²⁵ Though the scientist assures that Data would be reassembled with his knowledge and memories intact, Data remains convinced that the scientist does not have the sufficient expertise to reassemble him in any manner that allows him to retain the substance and flavour of his experiences but instead reduce them to mere facts; consequently, he refuses the request. The scientist then turns to the Starfleet Command to have them order Data to comply. After being advised that the only way for him to evade the order is to leave the Starfleet, Data submits his resignation.

In response to Data's resignation from the Starfleet, and still convinced that no human-made artefact can be regarded as some sort of legal person, the scientist asks the Starfleet court to confirm that Data is nothing more than the property of the Starfleet, obviously with no right to refuse to be disassembled or resign:

MADDOX: You are endowing Data with human characteristics because it looks human. But it is not. If it were a box on wheels, I would not be facing this opposition. (...)

MADDOX: Data must not be permitted to resign.

PICARD: Data is a Starfleet officer. He still has certain rights.

MADDOX: Rights! Rights! I'm sick to death of hearing about rights! What about my right not to have my life work subverted by blind ignorance?

PHILLIPA: We have rule of law in this Federation. You cannot simply seize people and experiment with them to prove your pet theories.

PICARD: Thank you.

MADDOX: Now you're doing it. Data is an extraordinary piece of engineering, but it is a machine. If you permit it to resign it will destroy years of work in robotics. Starfleet does not have to allow the resignation.

Data's commanding officer, Jean-Luc Picard, defends Data's right to self-determination before the court and argues that the legal status of Data cannot be reduced to that of property. In the hearing, the scientist argues that Data is nothing more than some form of advanced machinery. According to the scientist, though Data has humanlike physical features and is equipped the abilities that allow him to enter social interactions, he is ultimately nothing more than the combination of human-made software running human-made hardware. Interestingly, Captain Picard does not dispute the scientist's claims.

²⁵ Robert Scheerer, *Star Trek: The Next Generation – Season 2, Episode 9: The Measure of a Man* (1989).

Instead, he argues that the building blocks of Data cannot be considered relevant when answering the question of Data's legal status, considering that human beings are made from the building block of their parents' DNA but are not considered their property. Captain Picard then discusses the question of Data's sentience and convinces the court that Data satisfies two of the three criteria of sentience, intelligence, and self-awareness. He then compels the scientist to admit that the third criterion, consciousness, is one of the nebulous concepts whose elements cannot be clearly defined to decide whether Data satisfies it. On that account, Captain Picard submits that categorising intelligent, self-aware, and possibly sentient artificial entities like Data as mere property would mean to sanction slavery, with androids as the race of disposable creatures at the service of Federation, founded on the principle of personal liberty and freedom. Ultimately, the court finds in Data's favour and admits that he has the right to choose whether or not to participate in these experiments

In one episode of another series in the franchise, *Star Trek: Voyager*, titled *Author Author*, the holographic Doctor of another starship, the USS Voyager writes a holo-novel about himself called *Photons be Free* and submits the rough draft of that work to some publisher.²⁶ Though the Doctor decides to make some revisions to the novel not to damage his friends' reputations, he is informed that the publisher disseminated the novel without the Doctor's permission. The publisher defends the decision to publish the novel by arguing that the Doctor has no rights as the author of the novel since holographic entities are not regarded as legal persons. The Doctor's commanding officer, Kathryn Janeway, calls for the Federation tribunal to decide on the issue of holographic rights and brings several crew members as witnesses to the Doctor's claim of personhood. The Arbitrator who presides over the tribunal is reluctant to declare that the holographic Doctor has legal personhood, but due to the humanlike development of the Doctor's personality since he was first activated, he also cannot rule that the Doctor is mere software. In the end, the Arbitrator decides that the legal definition of the artist can be expanded to include the Doctor, giving him the right to control his work but not he refuses to give a definite answer to the personhood question.

²⁶ David Livingston, *Star Trek: Voyager – Season 7, Episode 20: Author Author* (2001).

IV.1.2 Psychological Experiments

The occurrence of anthropomorphism is rather incremental: Different human-like attributes, behaviours, and features displayed by non-human entities invoke various levels of anthropomorphic responses. Above, it was hypothesised that possessing attributes and behaviours perceived to be exclusive to human beings would result in more substantial occurrence of anthropomorphism and that hypothesis is supported by comparing the recorded experiences of masters of robots that only have the characteristic of relative autonomy and those with specific capabilities associated with social agency.

Still, the recorded experiences examined in the last section fail to reveal the degree of anthropomorphic responses that social robots can invoke. However, the strength of these anthropomorphic response should be known for the accurate assessment of the legal implications of robot anthropomorphism. Are social robots anthropomorphised at lower levels, similar to how human beings talk with their cars but ascribe no human-like intelligence or sentiments to them? Are they intuitively approached like companion animals -pets- that human beings form emotional connections? Do they invoke anthropomorphic responses to the degree that they are seen as our equals?

One of the first psychological experiments on the subject, performed by Rosenthal and others, illustrated the extent and strength of human beings' tendency to anthropomorphise robots that possess specific capabilities associated with social agency. In that experiment, participants' neurological responses while they were shown videos of one social robot (Pleo, an entertainment robot in the shape of a baby dinosaur) being mistreated and shown affection were compared with the same participants' neurological responses when they were shown videos of the identical behaviours performed toward human beings.²⁷ The neurological responses of participants when they were shown videos of the Pleo robot were measured to be similar to the responses invoked when they were shown videos of human beings.²⁸

Kate Darling, depending on the observations from the workshop she led, affirms that human beings are reluctant to mistreat robots with which they interact. In Darling's workshop, the participants were each assigned a robot partner, and they were initially asked to name these robot partners and then to spend some time with these robots; subsequently, they were asked to 'torture' and 'kill' robot partners assigned to them.

²⁷ Astrid Rosenthal-von der Pütten and others, 'An Experimental Study on Emotional Reactions Towards a Robot' (2012) 5 *International Journal of Social Robotics* 17, 19-21.

²⁸ *ibid* 29-32.

Darling notes that most of the participants in the workshop hesitated to do so, and the group who were given backstories about their robot partners that cited 'humanlike' characteristics of them outright refused to perform these actions.

In another experiment, dated 2019, Horstmann and others further explored the extent and strength of the human tendency to anthropomorphise robots that have the capabilities associated with the social agency. That study centred on testing the human participants' willingness to turn off their robot partners who were begging for their lives-pleading not to be switched off. The experiment's premise was based on the 1996 research on the Media Equation Theory by psychologists Clifford Naas and Byron Reeves.²⁹

The theory of Naas and Reeves was based on the evaluation of the research regarding the fact that human beings tend to be polite to computers. Through various stages of extensive observations, Naas and Reeves first established that human beings approach computers with male voices differently from those with female voices, and regard humanoid faces on screens as encroaching on their personal spaces. Interpreting these findings, they theorised that human beings' 'interactions with computers, television, and new media are fundamentally social and natural, just like interactions in real life'.³⁰ In other words, according to the Media Equation Theory, human beings are prone to regarding interactions with multimedia as social interactions, just as interactions between human beings.

In effect, the Media Equation Theory indicates that anthropomorphism is not necessarily prompted by physical features or even humanlike behaviours of non-human entities but also by their communicative interactivity. In other words, according to the Media Equation Theory, human beings regard interactions with multimedia as social interactions. Taking the Media Equation Theory as their starting point, Horstmann and others hypothesise that since social robots display communicative (and social) interactivity at higher levels than any other form of multimedia, human beings would tend to innately regard their interactions with social robots as 'especially socially' and therefore would be inclined to perceive social robots as 'living social beings'.³¹

²⁹ Byron Reeves and Clifford Naas, *The Media Equation* (CUP 1996) 15.

³⁰ *ibid* 5.

³¹ Aike C Horstmann and others, 'Do a Robot's Social Skills and Its Objection Discourage Interactants from Switching the Robot Off?' (2018) 13 PLOS ONE 1, 2.

Horstmann and others expected that human participants would be reluctant to turn off robots they just interacted with when these robots display the specific capabilities associated with the characteristic of social agency and provide emotional objections to being switched off; as human beings do not often 'power off' their social interaction partners in the ordinary course of everyday life.³² Horstmann and others' experiment was conducted with human participants who interacted with purpose-built social robots. When human participants were told to switch off the robots they just interacted with, half of them found their robot begging them with comments like, 'Please do not switch me off' or 'I'm afraid of the dark!'³³ Out of that half, one-quarter refused to switch their robot partners off, while the other quarter took three times longer to decide whether or not to switch their partners off than the remaining half of participants who did not hear such pleas from their robot partners.³⁴

Horstmann and others' study demonstrates that the Media Equation Theory applies to interactions between human beings and social robots; and that experiment also establishes human beings have the tendency to apply social norms when they are interacting with social robots – even though they know at conscious levels that social robots cannot experience humanlike feelings and sentiments.

Another similar experiment, performed by Bartneck and Hu is the roboticization of Milgram experiment. In original Milgram experiment, human participants were asked to administer electric shocks (that they did not know were fake) to 'learners' (played by research assistants). The electric shocks were gradually increased to levels that would have been fatal if they were real.³⁵ In Bartneck and Hu's version, instead of human beings, the 'learners' were social robots that could move their arms and emulate facial expressions that are associated with various human emotions on their faces.³⁶ In that experiment, human participants initially showed hesitation to administer higher voltages to social robots, but the experimenters' urges were enough to make all of them to continue increasing the electric shock until the maximum voltage was reached.

³² *ibid* 2.

³³ *ibid* 15.

³⁴ *ibid* 16.

³⁵ Stanley Milgram, 'Behavioral Study of Obedience' (1963) 67 *Journal of Abnormal and Social Psychology* 371, 375.

³⁶ Christoph Bartneck and Jun Hu, 'Exploring the Abuse of Robots' (2008) 9 *Interaction Studies* 415, 418.

In Milgram's original experiment, only 40% of the participants administered the deadly level of electric shock, whereas all the participants administered the maximum voltage with robots.³⁷ Bartneck and Hu's experiment shows that human beings have fewer concerns with mistreating robots in comparison with those they have mistreating human beings, though it took human beings time and several 'prods' by experimenters to override their emotional queasiness intellectually and harm robots that they knew was no more soulful than radio receivers. Accordingly, the results of the experiment in Bartneck and Hu's study can also be said to reinforce the suggestion that robot anthropomorphism should be regarded as an automatic cognitive response prompted by the specific capabilities of social robots that are associated with the characteristic of social agency.

Considering that human beings, in the ordinary course of everyday life, do not go through any moral struggles when switching off electronic devices, it is submitted that the illusion of mutual relating created by social robots, thanks to the characteristics of relative autonomy and social agency, strengthens the occurrence of the phenomenon anthropomorphism. Both recorded experiences and experimental research on robot anthropomorphism suggest that although human beings empathise with other human beings at higher levels than they do with robots, they nonetheless empathise with robots at substantial levels and project their human feelings and sentiments as a result of the phenomena of anthropomorphism. Social robots are no closer to the socio-psychological complexity of human beings than single-celled organisms or basic machines. There are nonetheless meaningful differences between how human beings interact with them. The next section of the present chapter focuses on that fact and explores the personal, economic, and social risks associated with robot anthropomorphism and then considers whether the intuition that it is wrong to mistreat robots is justified.

³⁷ *ibid* 420; Milgram (n 35) 371.

IV.2 Undesirable Outcomes

IV.2.1 Loss of Genuine Human Connections

Because of the capabilities of social agency, human-robot social interactions and interpersonal interactions have similar effects on human psychology. Though that similarity from the innate anthropomorphic tendencies of human beings, it is enough for the satisfaction the most basic social needs. Then, as social robots become more affordable, human beings who face loneliness or social alienation can be expected to turn to social robots as interaction partners. .

The concern here is that human beings may abandon pursuing any form of genuine human connection and ultimately altogether lose their capability to develop and sustain human relationships since human-robot social interactions do not have difficulties and intractable demands associated with interpersonal relationships.³⁸

Sherry Turkle notes that increasing numbers of human beings through interactions with social robots would eventually result in the breakdown of social cohesion and the loss of the sense of solidarity that helps maintain the social order:

I find people willing to seriously consider robots not only as pets but as potential friends, confidants, and even romantic partners. We don't seem to care what their artificial intelligences 'know' or 'understand' of the human moments we might 'share' with them (...) the performance of connection seems connection enough.³⁹

According to Turkle, as social robots are easier to engage with than human beings, human-robot social interactions may encourage human beings to avoid interacting with their friends and family.⁴⁰ Turkle argues that since the performance of social interaction is sufficient to satisfy the basic social needs of human beings, the ease of engaging with social robots might make human-robot social interactions preferable over interpersonal interactions.

Even though social robots are devised to make human beings they interact with feel understood at personal and social levels, they cannot understand the emotional and mental situation of any human being, including their human interactants. The characteristic of social agency does not allow social robots actually to care about human beings. The present study assesses that for any entity to be capable of caring for another, that entity must be able to determine its priorities.

³⁸ Matthias Scheutz, 'The Inherent Dangers of Unidirectional Emotional Bonds between Human Beings and Social Robots' in Patrick Lin, Keith Abney and George A Bekey (eds), *Robot Ethics: The Ethical and Social Implications of Robotics* (MIT 2014) 214.

³⁹ Turkle, *Alone Together* (n 17) 9.

⁴⁰ *ibid* 7.

Taking the term *priority* to refer to various value judgements and ultimate objectives that motivate the activities of autonomous entities, the present study finds that the priorities of robots are determined or delineated by their prescribed functions. No robot on the market can decide its priorities and genuinely care about human beings.

Despite the strength of anthropomorphic responses invoked by social robots and the illusion of mutual caring created thanks to these responses, the independent activities of social robots are also defined by the specified roles assigned to them. These roles, in turn, are determined by the interests of some human beings, those of robots' manufacturers or masters.⁴¹ Then, it can be argued that, unlike human relationships, human-robot social interactions are not founded on shared concerns, consequences, or common responsibilities. In other words, human-robot social interactions do not facilitate human beings' integration and inclusion within society.⁴² In that regard, some authors in the literature question the utility of human-robot social interactions, and they discuss whether the manufacturing of social robots should be subjected to legal restrictions.⁴³

Even though social robots are utilised for applications such as social interaction partners, the phenomenon of robot anthropomorphism and the framing of social robots as companions do not necessarily mean that human relationships will be replaced. On the contrary, some current applications show that social robots can contribute to the resolution of communication problems between human beings and thereby increase the strength of genuine human connections.⁴⁴ For instance, the NAO, a minuscule humanoid robot, is used for working with children with autistic spectrum disorders and is found to be effective in establishing eye-to-eye contact or basic social interactions, helping bridge communication gaps between teachers or parents and the children.⁴⁵ In one study to which Turkle herself contributed, it was found that Paro, the baby seal robot inspires conversation among nursing home residents when placed in common living areas.⁴⁶

⁴¹ Ibid 282

⁴² Turkle, *Alone Together* (n 17) 239; Sinziana M Gitiu, 'The Roboticization of Consent' in Ryan Calo, A Michael Froomkin and Ian Kerr (eds), *Robot Law* (Edward Elgar 2016) 207.

⁴³ Glenda Shaw-Garlock, 'Gendered by Design: Gender Codes in Social Robotics' in Marco Nørskov (ed), *Social Robots: Boundaries, Potential, Challenges* (Taylor & Francis 2018) 218.

⁴⁴ Diana Marina Cooper, 'The Application of a "Sufficiently and Selectively Open License" to Limit Liability and Ethical Concerns Associated with Open Robotics' in Ryan Calo, A Michael Froomkin and Ian Kerr (eds), *Robot Law* (Edward Elgar 2016) 173.

⁴⁵ Aida Amirova and others, '10 Years Of Human-NAO Interaction Research: A Scoping Review' (2021) 8 *Frontiers in Robotics and AI*.

⁴⁶ Kidd, Taggart, and Turkle (n 12).

Indeed, applications of social robots in health, caretaking, and special needs education show that social robots can be catalysts for forming new human relationships and strengthening existing ones when used rather than replacing them. As Darling sums up, though interactions with social robots can indeed replace human relationships, they can also facilitate communication between human beings; for example children and their teachers, doctors, and parents, presenting valuable supplements to human interactions.⁴⁷ In an interview, Cynthia Breazeal, leader of the team that created the first social robot, Kismet, also emphasises that social robots are meant to supplement human relationships rather than replace them and, therefore, they 'should be designed to support human empowerment'.⁴⁸

Therefore, the present study concludes that no clear case exists for categorically treating enhanced anthropomorphic responses invoked by social robots as problematic for maintaining authenticity in human relationships. Furthermore, there is no basis for introducing any general overarching legal constraint on the production of social robots. The manner in which social robots are utilised is critical here; supplementing human relationships rather than replacing these relationships appears to be crucial for helping drive the design of social robots in socially beneficial directions.

IV.2.2 Diminution of Privacy and Solitude

Even though individual interests are optimally fulfilled when human beings are living together in society, some of them, such as those in personal development and self-reflection, can only be fulfilled when one is alone, and isolated from social life. In that regard, it is assessed that the legal values protected by privacy rights, at the core, defines values ascribed by society to the moments that human beings are separate from society.⁴⁹

Though the concept of privacy is much older, the present study submits that national legal systems began to safeguard the concept and values it connotes directly and explicitly only in the 19th century. The developments regarding the legal protection of privacy rights were in response to the emerging threats posed to the integrity of these moments by technological developments and business inventions of the time, such as photography and yellow journalism.⁵⁰

⁴⁷ Darling, 'Who's Johnny?' (n 13) 177.

⁴⁸ Jedidiah Bracy, 'The Future of Privacy: My Journey Down the Rabbit Hole at SXSW' (*Privacy Perspectives*, 20 March 2015)

⁴⁹ Daniel J Solove, 'Conceptualizing Privacy' (2002) 90 California Law Review 1082, 1153-1154.

⁵⁰ Samuel D Warren and Louis D Brandeis, 'Right to Privacy' (1890) 4 Harvard Law Review 193, 196-197.

Since the inception of the legal protection of privacy values, the development of privacy rights has been intertwined with technological developments, and over the past two hundred years, it has become nearly impossible to determine the precise boundaries of these rights.⁵¹ Admittedly, in the present day, information security and privacy rights have already come to encompass the protection of human beings' personal spaces, property, and communications, as well as their freedom in determining the direction of intellectual, decisional, and behavioural aspects of their development.⁵² Legal values protected by privacy rights can be understood to include self-determination, secrecy, and solitude. Consequently, privacy rights protect one's freedom to decide the extent their life is made social. Interactions between privacy rights and technology are critical because of the threats of violation to these legal values posed by emerging technologies.

Most recently, the proliferation of the Internet was a milestone for the development of privacy rights. In response to the threats posed to privacy values by the affordances associated with the Internet, 'information protection' was included within the scope of privacy rights to ensure that the collection, use, and dissemination of personal information is under the control of persons on whom the information is about. The notion of information security emerged as part of privacy rights.⁵³ Significantly, the EU Charter of Fundamental Rights distinguishes between the right to privacy⁵⁴ and the right to data protection,⁵⁵ safeguarding both rights.

Another EU legal instrument, the General Data Protection Regulation, introduced in 2016, harmonises the national rules of the Member States on the subject matter, and ensures that individuals themselves have control observed how their personal information is utilised.⁵⁶

Though protections afforded to individual interests in information security and privacy have continuously expanded, legal values protected by privacy rights have changed little. The present study defines the primary purpose-values of privacy rights as 'to avoid interference with natural curiosity, introspection, and self-determination'.⁵⁷

⁵¹ Ruth Gavison, 'Privacy and the Limits of Law' (1980) 89 Yale Law Journal 421, 423.

⁵² Woodrow Barfield and Ugo Pagallo, *Advanced Introduction to Law and Artificial Intelligence* (Edward Elgar 2020) 78.

⁵³ David H Flaherty, *Protecting Privacy in Surveillance Societies* (UNC 1989) xiv.

⁵⁴ Charter of Fundamental Rights of the European Union [2012] OJ C326/95, Art 7.

⁵⁵ *ibid*, Art 8.

⁵⁶ Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the Protection of Natural Persons with Regard to the Processing of Personal Data and on the Free Movement of Such Data, and Repealing Directive 95/46/EC OJ [2016] L119/1 (General Data Protection Regulation).

⁵⁷ Ryan Calo, 'People Can Be So Fake: A New Dimension to Privacy and Technology Scholarship' (2010) 114 Penn State Law Review 809, 843-845.

Like the Internet before them, robotic technologies reveal new threats to values protected by information security and privacy rights in light of robots' ability to sense, process, record and broadcast what goes on around the environments in which they operate.⁵⁸ The broad range of sensors (cameras, laser and sonar range finders, GPS and so on) that they can be equipped with, together with both the variety in their shapes and the range of their physical abilities, presents some robots as unique surveillance.⁵⁹

Some robots, such as drones, have found significant uses in military and law enforcement for surveillance purposes. In the present day, some drones can stay aloft for days, navigate across landscapes autonomously, and stake out particular locations for more extended periods than any human being could, without getting detected.⁶⁰ The use of such types of robots are not restricted to security authorities of sovereign states. Autonomous aerial vehicles are freely available for purchase. In other words, legal persons -civilians- can utilise autonomous vehicles that they legally acquired for legitimate purposes such as securing their property and for illicit purposes such as voyeurism and casual trading.⁶¹ In short, the uncontrolled use of autonomous vehicles increases the likelihood of ubiquitous or mass surveillance and weakens the protection of privacy afforded to the individuals.⁶² Thanks to the capabilities associated with the robotic characteristic relative autonomy alone, some physical robots who are equipped with enhanced sensors make unprecedented levels of direct surveillance possible.

There is no shortage of legal instruments against the threat of direct surveillance. Prominently, the European Convention on Human Rights ['ECHR'] prohibits the invasions of privacy by both the public authorities and private entities⁶³ and the European Court of Human Rights has repeatedly described any form of mass surveillance activity as unlawful invasions to privacy rights in multiple judgements.⁶⁴

⁵⁸ Ryan Calo, 'Robots and Privacy' in Keith Abney, George A Bekey, and Patrick Lin (eds), *Robot Ethics: The Ethical and Social Implications of Robotics* (MIT 2014) 187.

⁵⁹ Lisa A Shay and others, 'Confronting automated law enforcement' in Ryan Calo, A Michael Froomkin and Ian Kerr (eds), *Robot Law* (Edward Elgar 2016) 242.

⁶⁰ Calo, 'Robots and Privacy' (n 58) 189; Cooper (n 44) 176.

⁶¹ Calo, 'Robots and Privacy' (n 58) 191.

⁶² Cooper (n 44) 180.

⁶³ Convention for the Protection of Human Rights and Fundamental Freedoms (European Convention on Human Rights, as amended) [ECHR], Art 8.

⁶⁴ *Roman Zakharov v. Russia* [2015] ECHR 1065; *Szabo and Vissy v Hungary* App No 37138/14 (ECHR, 12 January 2016); *Antovic and Mirkovic v Montenegro* [2017] ECHR 1068.

A similar approach is evident in standards established by UN legal instruments, although these standards are often criticised for being non-binding or non-enforceable.⁶⁵ The present study argues that the threats posed by direct surveillance to privacy rights that are protected by the law persists due to the difficulties in implementation of existing legal instruments, and not because of any inadequacy of the norms themselves. Apart from the threats to privacy posed by augmented surveillance opportunities facilitated by these capabilities of relative autonomy, specific capabilities of social robots that allow them to be anthropomorphic at unprecedented levels poses unique threats to privacy values. The present study addresses threats to privacy values that originate from the phenomenon of anthropomorphism in several categories. First of all, because of the substantial levels of anthropomorphic responses that social robots invoke, they can threaten the individual interest in solitude.

Solitude affords individuals the opportunity to self-reflect and allows them to decide on the direction of their self-development independently. However, thanks to the characteristic of social agency and strong anthropomorphic responses invoked by that characteristic, human beings automatically react to social robots as if they were other human beings.⁶⁶ In other words, when human beings are in the vicinity of social robots, they cannot behave as they would act on their own. Accordingly, the presence of social robots may reduce the opportunities for solitude, therefore may implicate privacy values.⁶⁷ In contrast with loneliness, solitude improves one's presence within society by facilitating self-exploration.⁶⁸ Considering that it is not only the privacy itself but the values that are protected by privacy norms that are threatened here, it is clear that this threat cannot be addressed by traditional privacy protections. However, in the near future, if social robots can identify the notion of personal space and are programmed to maintain socio-culturally acceptable distances from human beings, it is assessed that this threat is liable to disappear on its own.

⁶⁵ International Covenant on Civil and Political Rights (adopted 16 December 1966, entered into force 23 March 1976) 999 UNTS 171 (ICCPR) art 17; UN Human Rights Committee 'General Comment No 16' in 'Note by the Secretariat, Compilation of General Comments and General Recommendations adopted by Human Rights Treaty Bodies' (1988) UN Doc HRI/GEN/1/Rev.9 (Vol. I).

⁶⁶ Calo, 'People Can Be So Fake' (n 57) 843-845.

⁶⁷ Calo, 'Robots and Privacy' (n 58) 195.

⁶⁸ Julian Stern and Malgorzata Walejko, 'Solitude and Self-Realisation in Education' (2019) 54 *Journal of Philosophy of Education* 107, 109.

Secondly, thanks to their emerging position in society as independent social actors and preferred social interaction partners of human beings, social robots can be expected to gain access to historically private spaces such as bedrooms and living rooms and thus present increased dangers of surveillance.⁶⁹

Social robots that are capable of connecting to the Internet are assessed to 'create the possibility for unprecedented access to the interior of the house by hackers.'⁷⁰ Some robots designed for home uses already come with vast arrays of sensors. and some of these can even relay images, blueprints, and sounds across the Internet in real time.⁷¹ In other words, there are already functional artefacts capable of making extensive records of events in homes and communicating what they have recorded through the Internet. When any of these functional artefacts are hacked, hackers would have access to all the details of their victims' home lives.⁷² These details can include 3-D models of keys, for example, and victims can also be subsequently exposed to risks of physical intrusion.⁷³ In a recent study conducted by computer scientists at the University of Utah, a range of non-professional service robots currently on the market were tested against hacking, and most of them were found to be insecure and hijackable. The study revealed that hackers were able not only to eavesdrop on conversations but also to operate the robots.⁷⁴ However, it is submitted that the risk here can be significantly decreased by adopting stricter safety frameworks for the design and programming of robots.

Third, since social robots can interact socially with human beings and communicate with them at emotional and personal levels, new types of quantifiable sensitive personal information will inevitably be generated through interactions with social robots, increasing the gravity of potential breaches of information security.⁷⁵ By their very nature, social robots are able to employ emotional persuasion tools (for example fear or praise) to elicit confidence from human beings that they interact with, and they can encourage these human beings to reveal more about themselves than they would willingly and knowingly enter into databases.⁷⁶

⁶⁹ *ibid.*

⁷⁰ Calo, 'Robots and Privacy' (n 58) 192.

⁷¹ *ibid* 191.

⁷² Armağan Ebru Bozkurt-Yüksel, 'Robot Hukuku' (2017) 7 Türkiye Adalet Akademisi Dergisi 85, 96.

⁷³ Calo, 'Robots and Privacy' (n 58) 194.

⁷⁴ Tamara Denning and others, 'A spotlight on Security and Privacy Risks with Future Household Robots: Attacks and lessons' (11th International Conference on Ubiquitous Computing, Florida, October 2009)

⁷⁵ Cooper (n 44) 176.

⁷⁶ Darling, 'Extending Legal Protection to Social Robots' (n 10) 221; Daniel Dimov and Rasa Juzenaite, 'Privacy Concerns About Emotional Chatbots' (*InfoSec Institute*, 16 February 2018).

For example, just ten years ago, ElleGirlBuddy, a non- physical social robot tried to engage in social interactions with young adults and children to collect sensitive personal information which would then have been used for marketing purposes.⁷⁷

When interacting with social robots, thanks to the illusion of mutual caring created by the capabilities associated with the characteristic of social agency, human beings are motivated disclose their 'most intimate psychological attributes'.⁷⁸ As human beings manifest their innermost mental and emotional states through interactions with social robots, these states are can be transformed into digital data, creating new type of personal information that was previously impossible to collect 'whether through robot sensory equipment, or embedded as an expression of code'.⁷⁹ This way, intimate and non-physical experiences of human beings are transformed into quantifiable information for the first time, and this new type of highly sensitive personal information is vulnerable to the threat of cyber security infringements as much as any other information. Still, most social robots collect personal information in order to be able to connect with human beings on a more personal level and being overly proscriptive may circumvent the benefits of social robots.

That being said, the existing privacy legislation that is in place in all three national legal systems chosen for comparison, as well the EU's GDPR, espouses certain principles for ensuring the optimal level of data protection without hindering the provision of services that require data processing.⁸⁰ The principles espoused by the GDPR include data minimisation, purpose limitation, lawfulness, fairness, and transparency. Through these principles the GDPR aims to give some control over the collected data back to the subjects whose personal information constitutes the content of the data, as well as ensuring that the risk would be minimal if there is any breach of data security.

IV.2.3 Exploitation of Emotional Attachments

In 1976, Joseph Weizenbaum, after seeing how people interacted with ELIZA, the psychotherapist software he designed in 1960s, was compelled to warn human beings against being influenced by anthropomorphic machines and adopting the machines' and thereby their programmers' world views.⁸¹

⁷⁷ Ian R Kerr, 'Bots, Babes and the Californication of Commerce' (2004) 1 University of Ottawa Law and Technology Journal 285, 312-313.

⁷⁸ David Levy, *Love + Sex with Robots* (HarperCollins 2007) 22.

⁷⁹ Calo, 'Robots and Privacy' (n 58) 198.

⁸⁰ GDPR (n 56).

⁸¹ Joseph Weizenbaum, *Computer Power and Human Reason: From Judgment to Calculation* (Freeman 1976) 254.

Indeed, the breach of information security is not the only type of manipulation that warrants concern. Through social robots, human beings can be manipulated to do more than reveal their personal information or their vulnerabilities, as emotional attachments to these robots create additional potential exploitation sources.

To begin with, utilising their sociability and benefiting from the phenomenon of robot anthropomorphism, social robots, merely by expressing their unhappiness, may be used to manipulate human beings who are emotionally dependent to them to perform actions that they would not have performed otherwise. Matthias Scheutz refers to 'an admittedly futuristic sounding request of a robot dog to dispose of a real dog: "please get rid of this animal, he is scaring me, I don't want him around any longer"⁸² as an example of such cases. Since social robots have no individual interests to motivate them to exploit human beings' attachments to them unless they are programmed to do so, it is more likely that robot manufacturers will exploit attachments with robots for commercial purposes. For instance, a robot manufacturer may charge 'an exorbitant amount for a mandatory upgrade to a robot that someone's child or grandfather has become emotionally attached to' or 'a child's language-teaching robot' may 'have a vocabulary that is skewed toward specific products.'⁸³ Further, robot manufacturers might seek to manipulate the owner to buy products or to vote for a certain candidate.⁸⁴ The potential for exploitation infringes one of the fundamental interests that underlies most of the legal domains, the interest in the protection of the free will of humans.

Both the scope for emotional manipulation and the disclosure of sensitive personal information is assessed to have been at least partially addressed by the EU AI Act. The EU AI Act divides AI applications into several categories according to the risks they preserve, and on the high-risk applications that the Act seeks to ban altogether includes include those that deploy harmful 'subliminal techniques', 'exploit vulnerabilities' of specific groups, as well as those that are used by public authorities for social scoring purposes.

⁸² Scheutz (n 41) 16.

⁸³ *ibid.*

⁸⁴ Darling, 'Who's Johnny?' (n 13) 178.

IV.2.4 Robot Abuse

Most human beings respond to robots in a way that goes beyond thinking of them as mere functional artefacts. Even if the robot in question is not social (it does not have the specific capabilities associated with the characteristic of social agency), the relative autonomy is often sufficient to prompt human beings to empathise with and develop emotional attachments to it. Though no robot today is close to the intelligence and complexity of humans or animals, there are already significant differences in how humans interact with robots and with other artefacts. As the studies reviewed above identify, human beings show empathetic responses to the perceived pain of robots in similar manners to their responses to the perceived pain of other human beings. In other words, although robots do not feel pain or suffering, violent behaviours against them discomfort human beings.⁸⁵

It can be conjectured, therefore, that human beings have the intuition that torturing robots is morally wrong, even if they are just functional artefacts, abusive and violent behaviours toward robots are morally wrong, several authors suggest that such conduct may provide an effective method for deflecting violence away from human beings. For instance, the prominent psychologist Ronald Arkin raises the possibility that childlike sex robots could provide a safe outlet for human beings who are sexually attracted to children.⁸⁶ Conceivably, if the incidence of such behaviours against robots tended to reduce the incidence of such behaviours against living entities, the use of robots for these purposes might be thought to have important social benefits.⁸⁷

However, the probability that behaviours towards robots may have no significant effect or an ambiguous effect on the likelihood of someone performing their equivalent on living entities must be noted as well.⁸⁸ Ultimately, both sides of the debate lack empirical evidence and the collection of such evidence through scientifically sound, and ethically justifiable controlled psychological experiments appear to be nearly impossible:

Imagine, for instance, trying to design an experiment that could pass standard institutional review board scrutiny, if the objective was to test whether raping robotic sex dolls would make one more likely to perform such acts in real life to be a cathartic release for individuals with a proclivity for sexual violence, or have no noticeable effect on individual conduct.⁸⁹

⁸⁵ *ibid* 223.

⁸⁶ Kashmir Hill, 'Are Child Sex-Robots Inevitable?' (*Forbes*, 14 July 2014).

⁸⁷ John Danaher and Neil McArthur, *Robot Sex* (MIT 2017) 90.

⁸⁸ *ibid*.

⁸⁹ David J Gunkel, *Robot Rights* (Kindle edn, MIT 2018) ch 5, s 5.2, sub-s 5.2.4.

Thus, the present study supposes that -given the known empathetic responses to the abuse of robots- and the apparent intuition that torturing robots is morally wrong, mistreating robots may create a perception that it is acceptable to mistreat human beings - or at least other living entities - in the same way.

IV.3 Legal Protection of Social Robots

The following section analyses whether the intuition of human beings that it is morally wrong to abuse or torture social robots is justified. In the first instance, that intuition does not appear to be supported by most of the major normative approaches to ethics, namely consequentialism, deontology, and virtue ethics. Although social robots already undertake roles as independent actors in society and invoke substantial anthropomorphic responses from their human interactants, thanks to the capabilities associated with the characteristic of social agency, none of the criminal law systems imposes any punishments for human beings' offensive behaviours toward them.⁹⁰ We assess that the reason cannot be that the criminal law systems exclusively protect human beings since most criminal laws punish abusive behaviours toward animals,⁹¹ and some even prohibit desecrating particular inanimate objects such as national flags or religious artefacts.⁹²

IV.3.1 Moral Wrongness of Robot Abuse

To begin with, consequentialist theories of ethics hold that moral values' purpose is to describe the ideal human conduct. According to these theories, righteousness, the wrongness of human behaviours, or the moral worth of human actions, can be best determined by the consequences of these behaviours and actions. Right actions are the ones that have the maximum possible positive outcomes. Perhaps the best-known consequentialist theory is that of utilitarianism, where the righteousness of actions are taken to be those that maximise the 'happiness' or minimise the 'pain' of entities that are capable of subjectively experiencing and understanding these outcomes.

⁹⁰ Kamil Mamak, 'Should Violence Against Robots Be Banned?' (2022) 14 *International Journal of Social Robotics* 1057, 1061.

⁹¹ John Danaher, 'Robotic Rape and Robotic Child Sexual Abuse: Should They Be Criminalised?' (2014) 11 *Criminal Law and Philosophy*, 5; Darling, 'Who's Johnny?' (n 13) 179

⁹² John Hart Ely, 'Flag Desecration: A Case Study in the Roles of Categorization and Balancing in First Amendment Analysis' (1975) 88 *Harvard Law Review* 1482, 1485; Anita Dichter, 'Legal Definitions Of Cruelty and Animal Rights' (1978) 7 *Boston College Environmental Affairs Law Review* 147, 155.

Sentience -the capacity to feel, perceive, understand, and finally interpret the experience subjectively- unlike rationality, is not considered to be restricted to human beings. Bentham, one of the founders of utilitarianism, opposed cruelty to animals because animals have sentience. He argued that it was the ability to suffer that should determine the benchmark of how human beings treat other beings:

It may come one day to be recognized, that the number of legs, the villosity of the skin, or the termination of the *os sacrum*⁹³ are reasons equally insufficient for abandoning a sensitive being to the same fate. What else is it that should trace the insuperable line? Is it the faculty of reason, or perhaps, the faculty for discourse? [...] **[T]he question is not, Can they reason? nor, Can they talk? but, Can they suffer? Why should the law refuse its protection to any sensitive being?**⁹⁴ (*emphasis added*)

When robots are abused or tortured, in contrast to human beings or animals, they don't not feel any pain and suffering. Though there may be some damages on the robot, these damages are not experienced as equivalent to causing pain and suffering on living entities. Thus, it can be concluded that consequentialist theories do not provide any basis for the intuition that it is morally wrong to abuse robots. Deontology, another major normative approach to ethics, also does not appear to support the intuition about moral wrongness of robot abuse in the first instance.

According to Kant, the most prominent deontologist, the moral worth of any specific action is determined not according to the consequences, but according to the intention behind the action. If the action is impulsive and has no purpose, it will have no moral worth. Kant introduces the concept of the categorical imperative to evaluate the moral worth of the intentions behind the actions. The categorical imperative provides a test on maxims for determining whether the actions they refer to are right, wrong, or permissible. In its original formulation, the imperative states that one should 'act only according to that maxim whereby [one] can at the same time will that it should become a universal law'.⁹⁵

Deontological theories of ethics reject the notion that the moral worth of human beings' actions is determined by the utility of the consequences of these actions. Instead, these theories suggest that the intentions behind the behaviours of human beings determine the moral worth of said behaviours.

⁹³ i.e, a large, triangular bone at the base of the spine

⁹⁴ Jeremy Bentham and others, *An Introduction to The Principles of Morals and Legislation* (Clarendon 1996), 283.

⁹⁵ Immanuel Kant, *Groundwork of the Metaphysics of Morals* (CUP 1998) 30.

On the assumption that the purposes behind the actions of human beings are the existential causes of those actions, it can be argued that deontological theories measure the moral worth of actions by assessing whether their existential causes conform with specific moral standards. These moral standards are determined through the use of reason, meaning that without sufficient rational basis, no action can be categorised as morally wrong. *Prima facie*, it appears that the mistreatment robots would not be morally wrong since there are no explicit moral standards against violent behaviours toward robots. Moreover, since robots do not feel any pain or suffering, there seems to be insufficient justification for viewing such behaviours morally wrong from the deontological perspective.

Nevertheless, Kant's discussion of animal abuse offers a different perspective toward the violent behaviour of robots. Kant suggests that human beings are 'altogether different in rank and dignity from things, such as irrational animals, with which one may deal and dispose at one's discretion'.⁹⁶ but he also pointed out that human beings have duties, albeit indirect, towards animals:

So if a man has his dog shot, because it can no longer earn a living for him, he is by no means in breach of any duty to the dog, since the latter is incapable of judgment, but he thereby damages the kindly and humane qualities in himself, which he ought to exercise in virtue of his duties to mankind ... for a person who already displays such cruelty to animals is also no less hardened towards men.⁹⁷

It must be noted that although Kant is most closely associated with deontological ethics, the Kantian argument about animal rights is perhaps best assessed as an articulation of virtue ethics, the third major approach in normative ethics. In contrast to the deontological approach that stresses duties and rules, or the consequentialist approach that highlights the consequences of actions, virtue ethics emphasises virtue, character traits and not some result of the situational features in which one is embedded:

A virtue is an excellent trait of character. It is a disposition, well entrenched in its possessor—something that, as we say, goes all the way down, unlike a habit such as being a tea-drinker—to notice, expect, value, feel, desire, choose, act, and react in certain characteristic ways.⁹⁸

In line with the Kantian argument on animal rights, the virtue ethicist might consider the cruel treatment of animals to be wrong because it would impact on one's character in some adverse way.

⁹⁶ Immanuel Kant, *Lectures on Anthropology* (CUP 2012) 130.

⁹⁷ Immanuel Kant, *Lectures on Ethics* (CUP 1997) 212.

⁹⁸ Rosalind Hursthouse and Glen Pettigrove, 'Virtue Ethics' *The Stanford Encyclopedia of Philosophy* (Winter edn, 2018).

Drawing on this approach, robot ethicist Kate Darling has suggested that the 'Kantian philosophical argument for animal rights is that our actions towards non-human beings reflect our morality' and that 'this logically extends to our treatment of robotic companions'.⁹⁹ According to Darling, behaving violently towards 'very lifelike objects' not only reveals one's moral character, but can also change human beings and desensitise them to violent behaviours in other contexts.¹⁰⁰ As she puts it, 'if you're used to kicking a robot dog, are you more likely to kick a real dog'.¹⁰¹

Indeed, according to the virtue ethics approach, as illustrated in the context of the Kantian argument on animal rights, the mistreatment of a social robot would not be immoral because of the violation of a moral norm that protects the robot or due to its adverse consequences for the robot; but because it would shape one's moral character in the wrong way; per Coeckelbergh: 'Mistreating the robot is a vice.'¹⁰² Even though it is admitted that social robots cannot have any moral standing in their own right, that treating robots in morally questionable ways is likely to create and perpetuate the perception that it is acceptable to treat animals and human beings in a similar manner. The concern here is not that robots are moral victims, but instead because of their roles as participants in social interactions.

Given the empathetic responses to the mistreatment of robots and considering the intuition of human beings that abusive behaviours toward robots are abhorrent, it is submitted that regardless of whether robots can experience any pain or suffering, the abuse of robots is morally wrong. That assessment is supported, by way of analogy, through the statistical evidence pointing that human beings who mistreat animals are more likely to commit abusive and violent acts against other human beings and therefore show less empathy. For example, an analysis of the Chicago Police Department statistics pointed out that those arrested with violent offences often had prior charges of crimes against animals, and that, of those arrested for animal cruelty, 65% have later been arrested for violence against another person.¹⁰³

⁹⁹ Darling, 'Extending Legal Protection to Social Robots' (n 10) 220.

¹⁰⁰ *ibid* 224.

¹⁰¹ Kate Darling and Shankar Vedantam, 'Can Robots Teach Us What It Means To Be Human?' (*Hidden Brain*, 2017) .

¹⁰² Mark Coeckelbergh, 'Why Care About Robots? Empathy, Moral Standing, and the Language of Suffering' (2018) 20 *Kairos Journal of Philosophy & Science* 141, 145.

¹⁰³ Keri B Burchfield, 'The Nature of Animal Crime: Scope and Severity In Chicago' (2017) 64 *Crime & Delinquency* 1904, 1910.

Even though these statistics on animal cruelty do not provide decisive evidence for making accurate predictions about the impacts of animal abuse on one's character, they suggest that the ability to feel for, or empathise non-human entities might be a positive indication of one's own humanness, rather than merely some sort of delusion.

IV.3.2 Moral Wrongs and Criminal Law

In the previous section, it was argued violent behaviours toward social robots are morally wrong. Nevertheless, the premise that some behaviour is morally wrong - by and of itself - is not considered sufficient to justify the criminalisation of that behaviour. For instance, cheating in monogamous romantic unions, regardless of whether they are matrimonial or not, is widely considered immoral since it violates the ethical standards of honesty, fairness, and reciprocity. Still, adultery is not a crime in any European country, including those that are examined in the context of the present study. In that regard, the following section discusses whether it is feasible and possible for the criminal law systems to regulate abusive and violent behaviours against robots, regardless of whether such behaviours cause any extrinsic harm.

In a study that examines the topic of potential criminalisation of robotic rape and paedophilia, Danaher clarifies that criminalisation of abusive and violent behaviours against robots can be based on two straightforward premises:

- (1) It can be a proper object of the criminal law to regulate morally wrong conduct, even if such conduct has no extrinsically harmful effects on others.
- (2) Purely robotic acts of violence and sexual behaviours fall within the class of morally wrong but extrinsically harmless conduct that they could be proper objects of the criminal law to regulate.¹⁰⁴

The first premise can be regarded as the expression of the theory of legal moralism, that essentially holds that the criminal law may be used to prohibit certain behaviours based on the society's collective judgment of whether it is moral. This represents an alternative to legal liberalism which claims that laws can only intervene in order to protect the liberty of individuals. The clash between legal moralism and legal liberalism is popularised through Hart-Devlin debate. Hart summarises stances adopted by theories of legal moralism as follows:

¹⁰⁴ Danaher, 'Robotic Rape and Robotic Child Sexual Abuse (n 91) 76.

not only may the law be used to punish men for doing what morally it is wrong for them to do, but it should be so used; for the promotion of moral virtue by these means and by others is one of the ends or purposes of a society complex enough to have developed a legal system.¹⁰⁵

Hart does not essentially disagree with classicists' views that he summarises, to him, moral values that should be enforceable by means of the criminal law must be accompanied by close and imminent threats of any type of harm, and not just any future, indeterminate, uncertain risks. Without any proximate danger of harm, moral values are merely statements of the society's preferences, and these preferences cannot guide contents of the criminal laws. Though Devlin rightfully states that some shared preferences are beneficial for the continuation of society, according to Hart, in pluralistic liberal democracies, these preferences cannot be enforced by means of the criminal law. Accordingly, Devlin's legal moralism would be regarded as the enforcement of critical morality in Hart's terms. Devlin believed that since the society was maintained through certain shared moral values and as the law's business is to maintain the social order, the society every right to defend itself against any attack on that shared morality:

For society is not something that is kept together physically; it is held by the invisible bonds of common thought. If the bonds were too far relaxed the members would drift apart. A common morality is part of the bondage. The bondage is part of the price of society; and mankind, which needs society, must pay its price.¹⁰⁶

Devlin, therefore, asserts that for the sake of self-protection, the society is entitled to 'use the law to preserve morality in the same way as it uses it to safeguard anything else that is essential to its existence.'¹⁰⁷ Devlin's theory of legal moralism interprets these shared moral values as subjective 'sense of morality'. In other words, according to Devlin, the views of the average person living in the given society should be assessed to determine the content of that society's public and potentially legally enforceable values of morality.¹⁰⁸

The present study takes it for granted, in accordance with the discussion in the previous section, that mistreatment of robots would be regarded as contrary to shared moral values of society that can be enforced by the criminal law, because impacts of abusive and violent behaviours toward robots on general order and peace in society, such behaviours can be criminalised even if there are no extrinsic harms to determinate individuals.

¹⁰⁵ H. L. A Hart, *Essays in Jurisprudence and Philosophy* (Clarendon 1983) 248.

¹⁰⁶ Patrick Devlin, *The Enforcement Of Morals* (OUP 1965) 10.

¹⁰⁷ *ibid* 11.

¹⁰⁸ *ibid* 16.

It is hypothesised that the criminal law systems can accommodate the criminalisation of mistreating robots, as criminalisation efforts are not always based on the presence of extrinsic harm alone. Notably, animal protection legislation in selected legal systems does not appear to be exclusively based on the suffering of protected animals but instead appears to adopt a virtue ethics approach.

IV.3.3 Animal Protection Legislation

IV.3.3.a Germany

In Germany, since the Nazi Party's rise to power in 1933, animal cruelty has been taken very seriously under the law. Ironically, several prominent members of the Nazi Party, including Hitler, were staunch environmentalists. Consequently, animal welfare emerged as one of the prioritised items in the agenda of the Nazi government. Joseph Goebbels, the Nazi Propaganda Minister, reported in his diaries that Hitler's contempt for Judaism and Christianity largely originated from the distinction that these religions draw between the inherent value of human beings and that of animals; and he also stated that Hitler planned to ban the slaughter of animals altogether after the World War II.¹⁰⁹

Indeed, almost immediately after the Nazi Party's rise to power, the Reichstag began to pass laws for the regulation of animal slaughter. In August 1933, Germany became the first country in the world to ban vivisection.¹¹⁰ Meanwhile, one of the prominent members of Hitler's inner circle, Göring, banned commercial animal trapping, and imposed severe restrictions on hunting in the State of Prussia (of which he was the Minister-President) and also prohibited the boiling of live lobsters and crabs.¹¹¹

On November 24, 1933, the Reichstag enacted the Reich Animal Protection Act (Reichstierschutzgesetz). The Act introduced significant restrictions regarding human beings' behaviours toward animals, for instance, it prohibited the use of animals in filmmaking and other public events where there was any possibility of infliction of pain or injury to them, as well as feeding owls forcefully, and tearing out the thighs of living frogs.

¹⁰⁹ Louis Paul Lochner, *The Goebbels Diaries* (Award 1971) 679.

¹¹⁰ Arnold Arluke and Boria Sax, 'Understanding Nazi Animal Protection And The Holocaust' (1992) 5 *Anthrozoös* 1, 7.

¹¹¹ *ibid* 20.

Reichstierschutzgesetz was first in the world to extend any form rights to animals, providing legal protection of the animals regardless of the feelings of human beings toward them. It was also the first to abolish the distinction between domestic and wild animals, and extended legal protection to both as its subjects:

all living creatures that are in general language and biologically regarded as animals. In a criminal sense, there is no distinction between domestic and wild animals, higher or lower valued animals, or useful or harmful animals to human beings.¹¹²

The Reich Act stayed in place long after the collapse of the Nazi regime, until 1972, when it was superseded by the Animal Welfare Act (TierSchG), which took over its core provisions from the Reich Animal Protection Act. The most significant amendment to that Act was made in 2013.¹¹³ TierSchG defines the responsibility of human beings to treat animals as 'fellow creatures', and states that no person may cause an animal pain, suffering or harm without 'reasonable cause' – although the term is not defined in the legislation.¹¹⁴ Unlike its Irish counterpart, TierSchG legislation explicitly extends the protection provided to some invertebrates, including cephalopods and decapod crustaceans.¹¹⁵ Nonetheless, even with Germany's almost one-hundred-year-old progressive animal protection approach, the protection of animals seems to be limited to those that are able to demonstrate their pain and suffering through means that are similar to that of human beings. Insects or spiders are not protected in any way. This becomes even more obvious when the German restrictions on laboratory animals are considered. Accordingly, since 2010, in line with the EU legislation, lab experiments on monkeys have been tightened to a great extent - with only a few exceptions, experiments on great apes are nearly completely banned.

IV.3.3.b Italy

In Italy, the primary legislation regarding animal protection is the Law n. 189 of 2004,¹¹⁶ that amended the Penal Code¹¹⁷ and introduced new criminal offences that prohibit the mistreatment of animals, as well as their use in illegal fighting or unauthorised competitions and lists the acts which are considered acts of cruelty towards animals.

¹¹² Aslak Alkio, 'Animal Rights In The Third Reich' (2003) 2 (3) Kaltio (Online).

¹¹³ Tierschutzgesetz [TierSchG][Animal Welfare Act], 24 July, 1972, BGBl I at 1277, revised May 18, 2006, BGBl I at 1206, last amended by Art 105 of the Law of June 19, 2020, BGBl I at 3436, Arts 1, 4.

¹¹⁴ Ibid, Art 8.

¹¹⁵ Ibid, Art 9.

¹¹⁶ Legge, 20 luglio 2004, n 189, GU 31 luglio 2004, n 178,

¹¹⁷ Codice Penale [CP][Penal Code], Regio Decreto 19 ottobre 1930, n 1398, GU 26 ottobre 1930, n 251

These include killing animals cruelly or unnecessarily (CP Art 544-bis); cruelly or unnecessarily causing injury to an animal or subjecting the animal to torture, behaviour or overwork (CP 544-ter), and abandoning companion animals or animals whose natural behaviours were altered when they were in captivity (CP Art 727).

However, Italian law does not define what species can be categorised 'animal'; and therefore, it is not exactly clear whether the protection afforded extends to lower animals including fish and invertebrates.¹¹⁸ Then, it can be said to be more regressive compared to its German counterpart- as it does not recognise the sentience of even most human-like animals, let alone dissimilar animals. Like in other legal systems chosen for the purposes of the present study, Italian animal protection legislation does not seem to extend much protection to animals that do not show any human-like behaviours.

IV.3.3.c Ireland

In Ireland, the primary piece of legislation governing the welfare of animals is the Animal Health and Welfare Act 2013. The 2013 Act replaces the outdated provisions of the Protection of Animals Acts, 1911-1965. In the 2013 Act, protected animals are listed in Section 2 as follows:

'protected animal' means an animal—

- (a) kept for farming, recreational, domestic or sporting purposes in the State,
- (b) when it is in the possession or under the control of a human being whether permanently or on a temporary basis, or
- (c) that is not living in a wild state;

Clearly, Section 2 of the act is worded so as to include a broad range of animals. To be more precise, it can be said that the protected animals include cow, bullock, heifer, calf, steer, ox, equine, sheep, pigs (farming); horse, dogs, cats (sporting, recreational). Under the Act, other animals, such as avian animals or reptiles; as well as mammals that are unlike human beings (rats, rodents, gerbils) would only be afforded protection as long as they are laboratory animals or kept as pets. The Act prohibits the infliction of unnecessary suffering in the context of the duty to protect animals, a prohibition on animal cruelty, the proper feeding of animals, and a prohibition on animal fighting. Nonetheless, considering that the concept of 'unnecessary suffering' is quite vague, it is possible to state that laboratory animals, which are used for what would be considered as experiments

¹¹⁸ Italian case law provides that the animals that the relevant provisions of the penal code protect against ill-treatment include birds that can demonstrate discomfort or suffering in manners comparable to human beings, but not insects or mites. See, e.g., Cass, sez III Penale, sentenza, 29 aprile 2019, n 17691

necessary for progress of society, do not benefit from much protection anyway, further limiting the variety of animals protected under this legislation. Obviously, most members of the insect family, barring a few varieties of spiders that are kept as pets, are not to be protected under the Act.

In none of the chosen legal systems is animal protection legislation based on the notion of sentience. Although it is established that almost all animals feel pain or suffering when abused, tortured, and killed, the review of legal systems chosen for comparison in the present study demonstrates that the same degree of protection is not afforded to all animals. Rather, legislation often offers higher degrees of protection toward animals that are closest to human beings in appearance, or in how they respond to external stimuli. In other words, animal protection legislation appears to be based on the degree of anthropomorphism evoked by specific animals. For instance, the primates, which are closest to humans genetically and phenotypically, are the ones who benefit from the highest protection, followed by other mammals, followed by birds, and then reptiles. Insects, however, even though they are capable of feeling some degree of pain and suffering, do not seem to benefit from any protection whatever from abuse, torture or killing.¹¹⁹

Then, the protection afforded to non-human entities does not stem from their ability to feel pain, but rather from the similarities between their expressions of pain to human pain, or the general similarity of their appearance or behaviours and those of human beings. In other words, human beings frown upon animal abuse as long as they see some semblance between animals and human beings - if there is no semblance, then there is no protection.

Accordingly, it is possible to argue that the ban on animal abuse is not because of what animals feel. Instead, it is inspired by what human beings feel - and what they feel is closely linked to how much animals look like human beings and to what extent they can be anthropomorphised. Ostensibly, that can provide a formidable legal precedent to sanctioning robot abuse as well through criminal law.

¹¹⁹ The research shows that *drasophilia*, a species of fruit fly, can sense when it is injured, can learn to avoid noxious stimuli, and can differentiate between the degrees of adverse experience. Still, it cannot have appropriate responses to different degrees of injury, or employ mitigating behaviours, such as rubbing, limping, or guarding the injury. Nonetheless, *tobacco hornworm* can perform all these behaviours. Not only are insects not protected, some bird species seem to have been given protected status due to their tendencies to prey on insects and thereby keep the insect population in control. Lynne U Sneddon and others, 'Defining And Assessing Animal Pain' (2014) 97 *Animal Behaviour* 201, 204.

Regarding robot abuse, it is clear that robot anthropomorphism creates new social interests that demand recognition by the law. With that perception shift of robots, offensive behaviours toward robots draw public attention.

As the social dimensions of human-robot interactions get stronger, such behaviours toward robots can be expected to become more likely to effect social interests in general morality.¹²⁰ The present study suggests that criminal punishments can be used to protect social interests affected by offensive behaviours toward robots. Sanctions for these behaviours can be modelled after those foreseen for the cruel treatment of animals.¹²¹ Criminalisation of offensive behaviours toward robots cannot be precluded because no robot is capable of feeling any pain or suffering since not all criminal offences intend to uphold social interests in the protection of individual life. Indeed, there are plenty of victimless crimes in criminal law systems, such as recreational drug use, prostitution, and gambling, that are criminalised to vindicate social interests in general morals.¹²² It is worth mentioning that some proposals in the literature advocate prohibition of certain groups of robots that are designed to be subjected to offensive behaviours, such as 'child sex robots and sex robots designed to refuse advances'.¹²³

Summary

This chapter first explored anthropomorphism and examined the influence of robots' distinctive characteristics on the occurrence of the phenomena. Then, undesirable outcomes of robot anthropomorphism were discussed and whether there is any justification for the human intuition which dictates it is morally wrong to 'torture' social robots, even though they do not 'experience' anything. Finally, the selected legal systems' animal protection legislation was briefly reviewed to determine whether any parallels between animal protection legislation and potential protection for robots can be established.

¹²⁰; Roscoe Pound, 'A Survey of Social Interests' (1943) 57 *Harvard Law Review* 1, 25-26; D D Allan, Christoph Bartneck, and Andrew J Vonasch, 'The Doors of Social Robot Perception: The Influence of Implicit Self-Theories' (2021) 14 *International Journal of Social Robotics* 127, 128-129;

¹²¹ Kamil Mamak, 'Whether to Save a Robot or a Human: On the Ethical and Legal Limits of Protections For Robots' (2021) 8 *Frontiers in Robotics and AI*.

¹²² Louis Veneziano and Carol Veneziano, 'Are Victimless Crimes Actually Harmful?' (1993) 9 *Journal of Contemporary Criminal Justice* 1, 2.

¹²³ Danaher, 'Robotic Rape and Robotic Child Sexual Abuse (n 91) 35; Lily Frank and Sven Nyholm, 'Robot Sex and Consent: Is Consent to Sex Between a Robot and a Human Conceivable, Possible, and Desirable?' (2017) 25 *Artificial Intelligence and Law* 305, 320-32; Delphine DiTecco, 'New Technology, Same Old Stigma: An Analysis of Feminist Discourses and Sex Work Stigma in Sex Robot Media' (MA, Carleton University 2020) 22.

Through the comparison of animal protection legislation of chosen legal systems and different degrees of legal protection afforded to various animals, it is shown that when protection is afforded to animals, it is generally done in accordance with their mental complexity and similarity of the behaviours they demonstrate to those demonstrated by human beings - rather than just their ability to feel pain. The display of mental complexity seems to be why animals are protected, and the display of mental complexity also equates to some form of closeness to human beings.

It submitted that, in conclusion, animal sentience and or harm to animals is not a necessary prerequisite for their protection by the criminal justice system. By extension, the absence of harm to robots should not be thought to preclude the criminalisation of robot abuse.